

Data Analytics

Data Preparation

Data Preparation

- Also known as:
 - Data Preprocessing
 - Data Wrangling
 - Data Munging

Data Preparation

- Treat missing values
- Treat outliers
- Feature engineering
 - Transform existing features
 - Add new features (*extracted* from existing features)

Missing Values in Numerical Columns

- Most ML algorithms cannot deal well with missing values!
 - Delete column/examples with missing values
 - Impute (i.e. fill in) missing values

Types of Missing Values (Rubin, 1976)

- MCAR (Missing Completely At Random)
 - The probability that a value is missing is independent of both observed and unobserved data.
 - Missingness is entirely random; the observed data are a simple random sample of the full data.
- MAR (Missing At Random)
 - The probability that a value is missing may depend on observed data, but not on the missing (unobserved) value itself, once the observed data are conditioned on.
 - Missingness can be fully explained by variables that are observed.
- MNAR (Missing Not At Random)
 - The probability that a value is missing depends on the unobserved (missing) value itself, even after conditioning on observed data.
 - The missingness mechanism cannot be ignored and must be modelled explicitly.

Types of Missing Values (Rubin, 1976)

- MCAR (Missing Completely At Random)
 - Example: A survey response is missing because a questionnaire page was accidentally lost during data entry.
- MAR (Missing At Random)
 - Example: Income is missing more often for younger respondents, but within each age group the missingness does not depend on the income itself.
- MNAR (Missing Not At Random)
 - Example: Individuals with very high or very low incomes are less likely to report their income, and this reluctance depends on the income value itself.

Treating Missing Values

- If more than 25-30% values are missing in a column then drop the column.
 - Cons: information loss
- Delete examples/rows with missing values
 - Cons:
 - information loss
 - may introduce bias if not MCAR

Treating Missing Values

- Impute missing values
 - Simple method: use mean, median, most frequent, constant
 - Better method: KNN (K-Nearest Neighbour)

Handling Outliers

- Drop rows with outliers
 - Cons: information loss
- Add a new binary feature to mark outliers
- Transform
 - Log transformation
 - Cubic root transformation

Encoding of Categorical Attributes

- Most ML algorithms require all data to be numerical.
- Two types of categorical features:
 - Ordinal
 - ['low', 'medium', 'high']
 - Encoding: 'low': 0, 'medium': 1, 'high': 2
 - Nominal
 - ['apple', 'banana', 'orange']
 - Solution: one-hot encoding
(see *Encoding of Categorical Features.ipynb*)

Encoding of Cyclical Data

- Use a sine and cosine transformations
- Example notebook:

[How to Prepare Cyclical Data.ipynb](#)

Rescale Features and Normalization

- Many ML algorithms assume features are on the same scale.
- Scikit-learn Scalers:
 - MinMaxScaler
 - RobustScaler
 - StandardScaler
- Normalisation
 - Normalise data rows (when many features have the same nature, e.g. word frequency in text mining)

MinMaxScaler

`MinMaxScaler(feature_range=(0, 1), *, copy=True, clip=False)`

- Transforms a column by scaling it to a given range, typically [0..1].

- For each value x in the column:

$$z = (x - \min) / (\max - \min)$$

where $\min$ and $\max$ are the minimum and the maximum values in the column, respectively.

RobustScaler

`RobustScaler(*, with_centering=True, with_scaling=True,
quantile_range=(25.0, 75.0), copy=True, unit_variance=False)`

- For each x in a column:
$$z = (x - \text{median}) / \text{IQR} = (x - Q2) / (Q3 - Q1)$$
- This Scaler removes the median and scales the data according to the quantile range (defaults to IQR: Interquartile Range).
- Use RobustScaler if you want to reduce the effects of outliers, relative to MinMaxScaler.

StandardScaler

`StandardScaler(*, copy=True, with_mean=True, with_std=True)`

- For each value x in a column:

$$z = (x - u) / s$$

where u is the **mean** of the column or **0** if `with_mean=False`, and s is the **standard deviation** of the column or **1** if `with_std=False`.

- StandardScaler results in a distribution with a standard deviation equal to 1 and a mean approximately 0.

Normalizer

Normalizer(norm='l2', *, copy=True)

- Applied to rows, not columns!
- Each row of the dataset with at least one non zero component is rescaled independently of other rows so that its norm (L1, L2 or L^{inf}) equals one.
 - The L1 norm is the sum of the absolute values of all values in the row.
 - The L2 norm is the square root of the sum of the squared values in the row.
 - The L^{inf} also know as max norm is equal to the maximum absolute value in the row.

Question

- In the case of having outliers do you apply both a scaler and log transformation or both and in which order?

Create New Features

- Create/extract new features from the existing ones.
- The encoding techniques considered above do this.
- Other techniques can be used to construct features from existing ones.
 - Domain dependent